

AS.360.111 Earth is a Planet in Space (There are 5,000 Others)

Special Opportunity for Undergraduate Learning (SOUL) course
Fall 2024

Instructor: Katie Bennett

Class Location: Olin 304

Class Times: Thursdays 9:30am-12:00pm

Class Dates: Oct. 24th, Oct. 31st, Nov. 7th, Nov. 14th, Nov. 21st

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Office: Olin 315

Office Hours: By appointment

Course Description

This course will take the cosmic perspective and explain how many familiar phenomena – from the changing seasons to the vastness of our oceans – have their origins in the very formation of the Solar System. We will explore the root causes of these phenomena for Earth, and then explore what happens when these phenomena are turned on their heads, as in the cases of other Solar System planets or exoplanets that orbit other stars.

The course will broadly be split into three modules: composition, dynamics, and the stellar connection. In each, we will seek to answer a single question: 1) What is the Earth like, and why? (*composition*), 2) How does the Earth interact with all the other planets and bodies? (*dynamics*), and 3) Does a planet need to orbit a Sun-like star in order for it to have life (*stellar connection*)?

Learning Outcomes

By the end of the course, you should:

1. Describe the process of Solar System formation.
2. Explain the discovery and characterization of exoplanets.
3. Evaluate how phenomena here on Earth (seasons, hurricanes, mountains) are dictated by Earth's astronomical origin.
4. Predict what exoplanets might be like under different initial conditions. (i.e, are seasons inevitable? How tall could the highest mountain in the Galaxy be?)
5. Solve algebra-based problems to gain a deeper intuition in the study of exoplanets.
6. Interpret figures and diagrams from scientific papers.
7. Synthesize complex information graphically using Excel, Python, or another software/language of your choice.

Expectations

Your participation in this course is critical. I expect you to attend class in person and be actively engaged during the class. This means speaking at least once during each class, and attempting all in-class problems.

This class will rely on discussions and participation. For this reason, it is an in-person course. However, I realize that there are many circumstances in which it is not feasible for you to be here in person. Therefore, if you are unable to attend a class in person, please let me know beforehand and I will set up a Zoom room for you to participate remotely.

There are no textbook reading assignments for this course. Where appropriate, I will recommend some external reading if you are interested in learning more.

Evaluations

Assignments are as follows:

1. Each week, I will provide you some data related to a topic of discussion (e.g., the sizes of all known exoplanets.) You will be responsible for plotting the data in order to learn how to effectively convey information visually. I will not necessarily tell you how to plot the data (e.g., I won't tell you what your x- and y-axes should be). You will have to decide what works best. There will be a total of four or five plotting exercises.
 - a. Once I assign an in-class plotting exercise, the assignment will be due at 5pm EST the following Wednesday, giving you one full week to complete the assignment.
 - b. Note that I expect you will be able to mostly finish these assignments in class and bring minimal work home with you.

Grades are S/U:

- *Satisfactory*: You have attended at least four classes, are an active participant (speak at least once per class and attempt all in-class math problems), and have submitted all assignments (4-5 total).
- *Unsatisfactory*: You have attended three or fewer classes, have not participated when in class, or have not submitted all assignments.

If you are unable to attend a class session, please let me know beforehand.

Late Submissions

No late assignment submissions will be accepted, unless you have individually reached out to me with a request for a deadline extension before the assignment is due. If you request an extension deadline, I will grant a two-day extension (unless extenuating circumstances require a longer extension).

Academic Integrity

The strength of the university depends on academic and personal integrity. In this course, you must be honest and truthful. Ethical violations include cheating on exams, plagiarism, reuse of assignments, improper use of the Internet and electronic devices, unauthorized collaboration, alteration of graded assignments, forgery and falsification, lying, facilitating academic dishonesty, and unfair competition.

You may work together on the in-class math problems and the plotting exercises, but you must generate your own plots to turn in. You may use a large language model (e.g., ChatGPT) to help

de-bug a problem with Excel/Python, but you may not use it to generate your plots or solve any in-class math problems.

Report any violations you witness to the instructor. You can also contact:

- For undergraduates: the associate dean of student conduct (or designee) by calling the Office of the Dean of Student Life at 410-516-8208 or via email at studentconduct@jhu.edu
- For KSAS Graduate Students; rseitz5@jh.edu

Students with Disabilities

Johns Hopkins University values diversity and inclusion. We are committed to providing welcoming, equitable, and accessible educational experiences for all students. Students with disabilities (including those with psychological conditions, medical conditions, and temporary disabilities) can request accommodations for this course by providing an Accommodation Letter issued by Student Disability Services (SDS). Please request accommodations for this course as early as possible to provide time for effective communication and arrangements.

For further information or to start the process of requesting accommodations, please contact Student Disability Services at Homewood Campus, Shaffer Hall #101, call: 410-516-4720 and email: studentdisabilityservices@jhu.edu or visit the website <https://studentaffairs.jhu.edu/disabilities/>.

Anxiety, Stress, and Mental Health

If you are struggling with anxiety, stress, depression, or other mental health related concerns, please consider visiting the JHU Counseling Center. If you are concerned about a friend, please encourage that person to seek out their services. The Counseling Center is located at 3003 North Charles Street in Suite S-200 and can be reached at 410-516-8278 and online at <http://studentaffairs.jhu.edu/counselingcenter/>.

Classroom Climate

Everyone here has the right to be treated with dignity and respect. Please join me in creating a welcoming and vibrant classroom climate. Note that you should expect to be challenged intellectually by me and your peers, and at times this may feel uncomfortable. Indeed, it can be helpful to be pushed sometimes in order to learn and grow.

If you ever have concerns in this course about harassment, discrimination, or any unequal treatment, or if you seek accommodations or resources, I invite you to share directly with me. I promise that we will take your communication seriously and seek mutually acceptable resolutions and accommodations. Reporting will never impact your course grade. You may also share concerns with the Earth & Planetary Sciences (EPS) department chair (Ben Zaitchik, zaitchik@jhu.edu), the EPS Director of Undergraduate Studies (Meghan Avolio, meghan.avolio@jhu.edu), the KSAS Assistant Dean for Diversity and Inclusion (Araceli Frias, afrias3@jhu.edu), or the Office of Institutional Equity (oiie@jhu.edu). In handling reports, people will protect your privacy as much as possible, but faculty and staff are required to officially report information for some cases (e.g., sexual harassment).

Family Accommodation Policy

You are welcome to bring a family member to class on occasional days when your responsibilities require it (for example, if emergency childcare is unavailable, or for the health needs of a relative). Please be sensitive to the classroom environment, and if your family member becomes uncomfortably disruptive, you may leave the classroom and return as needed.

Schedule

Date	Topic
Oct. 24 th	Course Overview; Solar System formation and introduction to exoplanets
Oct. 31 st	<i>Composition Part I:</i> The Solid Earth (interior, plate tectonics)
Nov. 7 th	<i>Composition Part II:</i> Oceans and the Atmosphere
Nov. 14 th	<i>Dynamics:</i> tides, seasons/obliquity, asteroids, solar interactions
Nov. 21 st	<i>Stellar Connection:</i> (star types, binary systems) and the search for life